

# Circuit lower bounds

A Lean-linked draft summary

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This note records the proved scope of the cutwidth program and its aggregate-gate extensions. The linked results are kernel-checked or direct specializations; conditional statements retain explicit unproved premises, and paper-only deductions are marked separately. Historical priority is a separate question; this is not a claim that every entry is a new literature record. Lean links resolve to files in the accompanying repository checkout.

## 1. The explicit Boolean family and the binary bounds

For  $n \geq 1$ , the fixed family is

$$f_n(b, x) = b \oplus E_{n-1}(x), \quad x \in \{0, 1\}^{n-1}.$$

Here  $E_m$  is the actual deterministic evaluator implementing the [Chattopadhyay–Liao sumset-extractor strategy](#) (Lemma 5.4 and the proof of Theorem 2), with the library’s conservative parameters. Schematically, it takes a majority of XORs of affine-correlation-breaker calls, whose seeds come from a sampler and whose advice encodes the enumeration indices. The total evaluator caps candidate enumeration; this cap is eventually inactive. The fresh bit  $b$  makes the family exactly balanced at every positive length.

One uniform evaluator and membership in  $\mathbf{P}$  are proved. Its polynomial degree and the onset of the asymptotic guarantee are enormous. The useful property is that, eventually, some  $K_n$  with  $\log_2 K_n = o(n)$  satisfies, for every  $P, Q \subseteq \{0, 1\}^n$ ,

$$|P|, |Q| \geq K_n \implies \{f_n(p \oplus q) : p \in P, q \in Q\} = \{0, 1\}.$$

*Lean:* [family definition](#), [uniformity and balancing](#), [sumset dispersion](#).

Write  $g$  for the number of gates and set

$$L = 1 + \frac{\pi}{3 \arccos((1 + 2\sqrt{2})/4)} = 4.5624976789 \dots$$

The notation  $g \geq an - o(n)$  abbreviates: for every  $\varepsilon > 0$ , all sufficiently large  $n$  require more than  $(a - \varepsilon)n$  gates. Boolean circuits are finite acyclic circuits with arbitrary depth, fanout and repeated slots. Every gate costs one;  $n$  includes the balancing bit.

| Model            | Signature and restrictions                                                                                                 | Bound for $f_n$    |
|------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------|
| Deterministic    | $\mathbf{B}_2$ : all sixteen binary Boolean functions; no further restriction.                                             | $g \geq Ln - o(n)$ |
| Nondeterministic | $\mathbf{B}_2$ on $n$ ordinary inputs and any number of existential witness inputs; the bound counts ordinary inputs only. | $g \geq Ln - o(n)$ |

*Core idea.* A narrow layout of the cubic core of a small circuit forces accepting rectangles excluded by the hard family; existentially projecting witnesses preserves the counting argument. The Gaussian edge-score layout supplies the coefficient  $L$ . *Lean:* [explicit binary theorem](#), [Gaussian and nondeterministic theorems](#). The nondeterministic specialization uses the checked [density and rectangle properties](#). The older coefficients 4 and  $41/9$  are weaker corollaries.

## 2. Unbounded gates: sparse and whole-basis results

**Finite aggregate signature.** Besides  $B_2$ , a special gate  $j$  may compute

$$h_j \left( \prod_i \phi_{ji}(y_i) \right), \quad \phi_{ji} : \{0, 1\} \rightarrow M_j, \quad h_j : M_j \rightarrow \{0, 1\},$$

where  $M_j$  is any finite commutative monoid. The slot maps and Boolean readout are arbitrary. Fan-in and placement are unrestricted, including connections between special gates. Let  $q$  count special gates and  $s_2$  count binary gates, so  $g = s_2 + q$ , and write  $B = s_2 + \sum_j \log_2 |M_j|$  for exact state capacity. This signature includes AND, OR, parity and modular counting; larger registers also cover symmetric predicates and finite weighted counters.

| Result           | Restriction                                                    | Bound for the same $f_n$     |
|------------------|----------------------------------------------------------------|------------------------------|
| Sparse extension | $D = q + \sum_j \lceil \log_2  M_j  \rceil = o(n)$ .           | $g \geq Ln - o(n)$           |
| Exact capacity   | None on the number or sizes of the registers.                  | $B \geq n - o(n)$            |
| Bounded states   | $ M_j  \leq r$ for fixed $r \geq 2$ ; no sparsity requirement. | $g \geq n / \log_2 r - o(n)$ |

*Sparse method.* Guess special outputs and retain their finite aggregate states across cuts, using outgoing-transition counting to pay only  $D$  extra bits. The condition  $D = o(n)$  is a circuit restriction, not an unproved hypothesis. *Lean:* [aggregate hardness](#).

*Capacity method.* Alice sends the aggregate of direct primary-input contributions to each gate; Bob evaluates the circuit, while rectangle hardness forces nearly  $n$  bits of total message capacity. Bounding each register by  $r$  states gives the gate-count corollary. *Lean:* [capacity and bounded-state bounds](#). For example,  $r = 3$  gives  $g \geq 0.6309297536 \dots n - o(n)$ .

**The full signed AND/OR/XOR basis.** For every finite arity  $r$ , allow either operation

$$b \oplus \bigoplus_{i=1}^r (a_i \wedge y_i), \quad b \oplus \bigwedge_{i=1}^r [y_i = p_i], \quad a_i, b, p_i \in \{0, 1\}.$$

Thus all affine Boolean gates and optionally complemented signed conjunctions cost one gate. This includes unbounded AND, OR and XOR and every binary Boolean operation. There is no sparsity, fan-in, depth or placement restriction. Define binary entropy by  $H_2(t) = -t \log_2 t - (1 - t) \log_2 (1 - t)$ , and let

$$h = H_2(1/4), \quad r = 1 - H_2(1/8), \quad C = \frac{3 - h + 3r}{2 + 2r} = 1.22148505965 \dots \quad \boxed{g \geq Cn - o(n)}.$$

*Core idea.* Shared primary inputs allow one affine restriction to simplify two conjunctions. A maximal pairing supplies an extra geometric saving; joint entropy handles two-variable summaries, and stronger bias handles wider conjunctions. This improves the retained coefficients  $1.19065368005 \dots$  and  $1.158760328 \dots$ . There are no unresolved hypotheses. The basis does not include  $\text{MOD}_3$  or majority. *Lean:* [signature](#), [one-gate embedding of  \$B\_2\$](#) , [shared primary controls](#), [joint information bound](#), [finite combination](#), [explicit-family bound](#).

### 3. A natural multi-output target: finite-field inversion

Fix  $n \geq 3$ , a field  $K$  of  $2^n$  elements, and any supplied linear coordinate isomorphism  $e : \mathbb{F}_2^n \rightarrow K$ . Define

$$I_e(x) = e^{-1}(e(x)^{-1}), \quad 0^{-1} = 0.$$

Put  $c = 1 - H_2(1/4)$ . A circuit computing *all  $n$  output bits* satisfies

$$g \geq \frac{3+2c}{2+c}n - \frac{4c}{2+c} = 1.54311234736\dots n - 0.34489877887\dots$$

This strengthens the retained theorem  $g \geq \frac{3}{2}n - 2$ . Every scalar Boolean gate has unit cost, including unbounded signed AND, OR and XOR; the basis includes all of  $\mathbf{B}_2$ . Depth, fanout, fan-in and gate placement are unrestricted. The bound concerns all coordinates together, not an individual coordinate or a circuit with field-arithmetic operations as gates.

**Three costs combine.**

- *Affine restrictions.* Every affine subspace on which all inverse coordinates are affine has at most four points. A zero-avoiding affine plane has a nonzero sum of reciprocals, obstructing larger affine restrictions. If  $h_2$  counts conjunctions with at least two distinct direct primary inputs, affine pairing therefore gives  $g + h_2 \geq 2n - 4$ .
- *Independent nonlinear components.* Every nonzero XOR of inverse coordinates is nonaffine in the primary bits. Modulo affine primary functions, conjunction outputs span all circuit outputs, so there must be at least  $n$  conjunction gates. If  $o$  of the distinct output gates are conjunctions, the remaining  $n - o$  output gates give  $g \geq 2n - o$ .
- *Information.* Inversion is a permutation, so the full primary-message vector is injective. A balanced nonliteral conjunction output cannot contain a direct primary literal; its primary summary is constant. Removing these  $o$  messages and accounting for the quarter-biased  $h_2$  messages gives  $n \leq g - o - ch_2$ .

The last two inequalities imply  $2g \geq 3n + ch_2$ . Eliminating  $h_2$  with the first proves  $(2+c)g \geq (3+2c)n - 4c$ . All inversion properties are proved directly; there is no supplied extractor, dispersion, or hardness hypothesis.

*Lean:* [finite inversion bounds](#), [exact constants](#), [reciprocal identities](#), [nonaffine components](#), [nonlinear-generator rank](#), [output information bound](#).

**Uniformity boundary.** The theorem accepts every supplied field and linear basis. This slice does not formalize a canonical choice for each  $n$ , or a uniform polynomial-time evaluator for that choice. The scalar family retains its separate checked uniform-P theorem.

**Source credit.** The reciprocal-sum geometry is classical; see Claude Carlet, *On the vector subspaces of  $\mathbb{F}_{2^n}$  over which the multiplicative inverse function sums to zero*, *Designs, Codes and Cryptography* 93 (2025), 1237–1254. The circuit deduction is separate; no literature-priority claim is made here.

## 4. Other explicit targets and graph consequences

The field-valued circuit results below allow *arbitrary* functions of at most two field symbols, including unary functions and constants. They impose no arithmetic-gate, linear-circuit or bilinear-circuit restriction. The finite base field may vary with the dimension. Size counts gates on field symbols, not bits, except when the base field is  $\mathbb{F}_2$ .

| Target                     | Function condition and dimensions                                              | Gate lower bound        |
|----------------------------|--------------------------------------------------------------------------------|-------------------------|
| Linear map $x \mapsto Mx$  | $N$ inputs and $N$ outputs; every square submatrix of $M$ is nonsingular.      | $LN - o(N)$             |
| Quadratic form $x^\top Mx$ | $N$ inputs, one output; every square submatrix of $M + M^\top$ is nonsingular. | $\frac{L+1}{2}N - o(N)$ |
| Field multiplication       | Degree- $m$ extension in any basis; $2m$ inputs and $m$ outputs.               | $(L+1)m - o(m)$         |

**Linear maps.** Explicit Cauchy matrices over  $\mathbb{F}_q$ , for primes  $q \geq 2N$ , satisfy the stated condition (total regularity). *Core idea.* Full-rank matrix blocks require about  $N$  signals across a terminal-balanced cut, opposing the small-circuit layout bound. *Lean:* [totally regular maps and Cauchy examples](#).

**Quadratic forms.** Explicit symmetric Hankel Cauchy matrices over prime fields  $q > 2N$  satisfy the polarization condition. The coefficient is  $(L+1)/2 = 2.7812488395\dots$ . *Core idea.* Vanishing mixed second differences on boundary rectangles force orthogonality through the cross-block matrix, giving a rank obstruction. *Lean:* [quadratic forms and Hankel Cauchy examples](#).

**Multiplication.** In particular, multiplication in  $\mathbb{F}_{2^m}$  in any basis requires  $5.5624976789\dots m - o(m)$  full-binary-basis gates. There are  $2m$  input bits, so the coefficient per total input bit is  $2.7812488395\dots$ . *Core idea.* Fixing one factor and averaging the resulting linear multiplication maps forces nearly full-rank blocks across a cut. *Lean:* [finite-field and binary multiplication](#).

All three results are unconditional for the stated target classes. The exact constants for the explicit Cauchy examples follow from the general theorems; their separately named explicit corollaries use the weaker rational constants  $41/9$  and  $25/9$ .

**Superconcentrators.** An  $N$ -superconcentrator has  $2N$  distinct designated input and output vertices and routes any equally sized sets of its  $N$  inputs and  $N$  outputs by vertex-disjoint directed walks. For finite directed multigraphs, with no acyclicity assumption, the checked bounds are

$$\#\text{edges} \geq (L+1)N - o(N); \quad \#\text{non-input vertices} \geq LN - o(N)$$

where only the second bound requires input indegree zero and indegree at most two everywhere. These are graph bounds, not additional gate signatures. *Core idea.* Routing forces  $N$  edges across a terminal-balanced cut; degree reduction then brings the graph under the improved cubic ordering theorem. *Lean:* [edge and vertex bounds](#).

## 5. Conditional statements and remaining boundaries

**Gaussian-band improvement: an open graph hypothesis.** For  $w > 0$ , assume [BandSubcritical](#) for every kernel decay  $q \geq 0$  with  $2q^2 < 1$  and every truncation radius  $R$ . It says, uniformly over all finite simple cubic graphs, that the expected fraction of vertices in large components of the edge-score band  $[-w, w)$  tends to zero as the allowed component size grows. Then  $B_2$  circuits for the same explicit  $f_n$ , with no further restriction, satisfy

$$g \geq [1 + (L - 1)e^{w^2/2}]n - o(n).$$

At  $w = 4/25$ , the coefficient is  $4.6083907382\dots$ , with a named rational weakening  $23/5 = 4.6$ . The implication is checked; uniform band subcriticality is not proved. *Core idea.* Small band components allow jumps through the Gaussian layout, reducing its frontier cost. *Lean:* [conditional explicit-family bounds](#).

**Average case: stronger assumptions on the target.** Suppose a Boolean family  $h_n$  is within  $\nu \geq 0$  of balanced on every coordinate rectangle whose two sides have size at least  $K_n$ , where  $K_n \leq n^a$  for a fixed  $a$ . For every fixed  $\varepsilon > 0$  and all sufficiently large  $n$ , every  $B_2$  circuit of size at most  $(4 - \varepsilon)n$  has uniform-input agreement at most

$$\Pr_x[\mathcal{C}(x) = h_n(x)] \leq \frac{1}{2} + 3\nu + 2^{-\varepsilon n/24}.$$

Thus any fixed advantage above  $3\nu$  requires  $4n - o(n)$  gates. The required quantitative rectangle balance has not been supplied for our concrete  $f_n$ ; exact global balance alone is insufficient. *Core idea.* Directional cut accounting controls rectangle multiplicities and hence prediction advantage. *Lean:* [average-case theorem](#).

**Paper deductions, not separately formalized results.** For the same  $f_n$ , the research notes additionally establish:

- $B_2$  plus  $q$  parity gates: if  $2q \leq (1/3 - \delta)n$  for fixed  $\delta > 0$ , then  $g \geq Ln - (4L - 3)q - o(n)$ ; the method is sharper aggregate-register accounting.
- The sparse aggregate bound survives a free invertible affine change of exactly  $n$  input coordinates; the method is affine invariance of sumset dispersion, without free internal affine gates.
- In the signed AND/OR/XOR basis,  $g \geq 2n - o(n)$  if every nonlinear gate has no direct primary-input slot; if instead no nonlinear gate has exactly one distinct direct primary input,  $g \geq (2 - h)n - o(n)$ . Both separate nonlinear restriction costs from input-information costs.

*Notes:* [aggregate transfer](#), [placement deductions](#). No improvement beyond  $5n$  for  $U_2$  is established here.

**Attribution.** The cutwidth architecture follows Ryan Williams’s private working note and Samuel Schlesinger’s circuit compiler. Gaussian-process antecedents include [Cs6ka–Gerencs6r–Harangi–Vir6g](#). Affine restriction and gatewise communication have classical antecedents in [Demenkov–Kulikov](#) and [Roychowdhury–Orlitsky–Siu](#); in particular the capacity row is not presented as a new communication technique. See the [source guide](#) for the graph, extractor and superconcentrator provenance and the distinction between borrowed ingredients and deductions developed in this project.